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Gut Microbiome Metagenomics Analysis (16S rRNA V3-V4)

Patient name	:	**XXXXX**	**PIN**	:	**XXXXXX**
Gender/ Age	:	**XXXXX**	**Sample number**	:	**XXXXXX**
Specimen	:	**Stool**	**Sample collection date**	:	**XXXXXX**
Referring Clinician	:	**XXXXX**	**Sample receipt date**	:	**XXXXXX**
			Report date	:	**XXXXXX**

Clinical history

To Identify the gut microbial profile composition.

Results Interpretation

Probiotic Biome

The probiotic biome consists of beneficial microorganisms that reside in the human gut, playing a vital role in overall health. These microbes support digestion, strengthen the immune system, and contribute to general well-being. Among the most used probiotics are *Lactobacillus* and *Bifidobacteria* [1].

Microbial Group	Abundance (%)
Akkermansia muciniphila	0
Bacillus subtilis	0
Bifidobacterium adolescentis	2.68
Bifidobacterium animalis	0
Bifidobacterium bifidum	0.18
Bifidobacterium breve	0
Bifidobacterium longum	0.37
Enterococcus durans	0
Heyndrickxia coagulans	0

Lacticaseibacillus casei	0
Lacticaseibacillus paracasei	0
Lacticaseibacillus rhamnosus	0
Lactiplantibacillus pentosus	0
Lactiplantibacillus plantarum	0.05
Lactobacillus acidophilus	0
Lactobacillus delbrueckii	0.01
Lactobacillus gasseri	0
Lactobacillus helveticus	0
Lactobacillus johnsonii	0
Lactococcus lactis	0
Leuconostoc mesenteroides	0
Levilactobacillus brevis	0
Ligilactobacillus salivarius	0.007
Limosilactobacillus fermentum	0
Limosilactobacillus reuteri	0
Streptococcus thermophilus	0.502

Pathogenic Biome

These microorganisms, including bacteria, viruses, and fungi, can disrupt the delicate balance of the human microbiome and negatively impact health. Understanding the dynamics of the pathogenic biome is essential for identifying and combating infectious agents that contribute to various diseases.

Microbial Group	Abundance (%)
Bacillus cereus	0
Bacillus cytotoxicus	0
Campylobacter jejuni	0
Citrobacter freundii	0
Clostridioides difficile	0.4

Clostridium botulinum	0.01
Enterococcus faecalis	0
Enterococcus faecium	0
Escherichia coli	0.03
Fusobacterium nucleatum	0
Haemophilus influenzae	0
Helicobacter pylori	0
Klebsiella pneumoniae	0
Listeria monocytogenes	0
Mycobacterium avium	0
Proteus mirabilis	0
Pseudomonas aeruginosa	0
Salmonella enterica	0.24
Shigella boydii	0
Shigella dysenteriae	0
Shigella flexneri	0
Shigella sonnei	0
Streptococcus pneumoniae	**0.01**
Vibrio cholerae	0
Yersinia enterocolitica	0

Test Methodology

DNA is extracted using QIAamp DNA Mini Kit. The extracted DNA is quantified using Qubit dsDNA High sensitivity Assay kit (Invitrogen, Cat# Q32854) (Qubit Fluorometer 3). The library is sequenced for 16S V3V4 amplicon sequencing in Novaseq X PlusIllumina platform.

Quality Check

Quality check on the raw Fastq files was done using FASTQC toolkit to check for base quality, base composition, and GC content. Quality filtering is performed to remove low-quality reads that may contain sequencing errors or low-confidence base calls. This helps to improve the accuracy of downstream analyses.

Paired-end Merging

Forward and reverse reads are stitched together using a join pairs module of QIIME2 with default parameters [12].

Pre-processing of Reads

De-replication was performed using dereplicate-sequences in VSEARCH for the identification of unique sequences. Chimeric sequences were filtered out using the UCHIME utility from VSEARCH using a de novo approach [13].

Operational Taxonomic Unit (OTU) Identification and Singleton Removal
Sequences having a similarity of 97% are grouped together using a closed reference method for classification of the whole data. Any OTU that has a count of 1 in a single sample (identified as singletons) are filtered out as these may have been created as experimental artifact.

Taxonomy Classification
One representative sequence from each OTU was picked up and are classified using RDP classifier against SILVA-138 database at 97% similarity. The taxonomy classification was done at phylum, class, order, family, genus, and species level [14]. Also, Kraken analysis against Mini kraken v2 database (Refseq: bacteria, archaea, viral) based approach to classify the organisms.

Alpha Diversity
Alpha diversity measures the diversity within an individual sample. It is measured in terms of Chao1 and Shannon index. Shannon index measures alpha diversity in terms of total taxa and their relative abundances and Chao1 measures the estimated richness of a sample in terms of how many different taxa are present.

Test Limitation and Disclaimer

The aim of this test is to provide physicians with an objective molecular diagnosis of the patient's gut microbial health. Depending on the result of this analysis, the physician may use it to improve the gut microbial health.

The results may vary depending on different factors such as hormonal changes, antibiotic intake, sexual activity, hygiene, etc.

The results may be affected by contamination of the sample at the time of specimen collection and/or during transportation. Stringent precautions have been taken to prevent cross contamination in the laboratory.

Data Summary:

Sample name	Total reads (R1+R2)	GC %	Read Quality	Data Size (in GB)
XXXXX	888.486000 K	55.80	90.45	0.26

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Appendix

List of complete genus and species level classification click here.

This report has been reviewed and approved by:

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